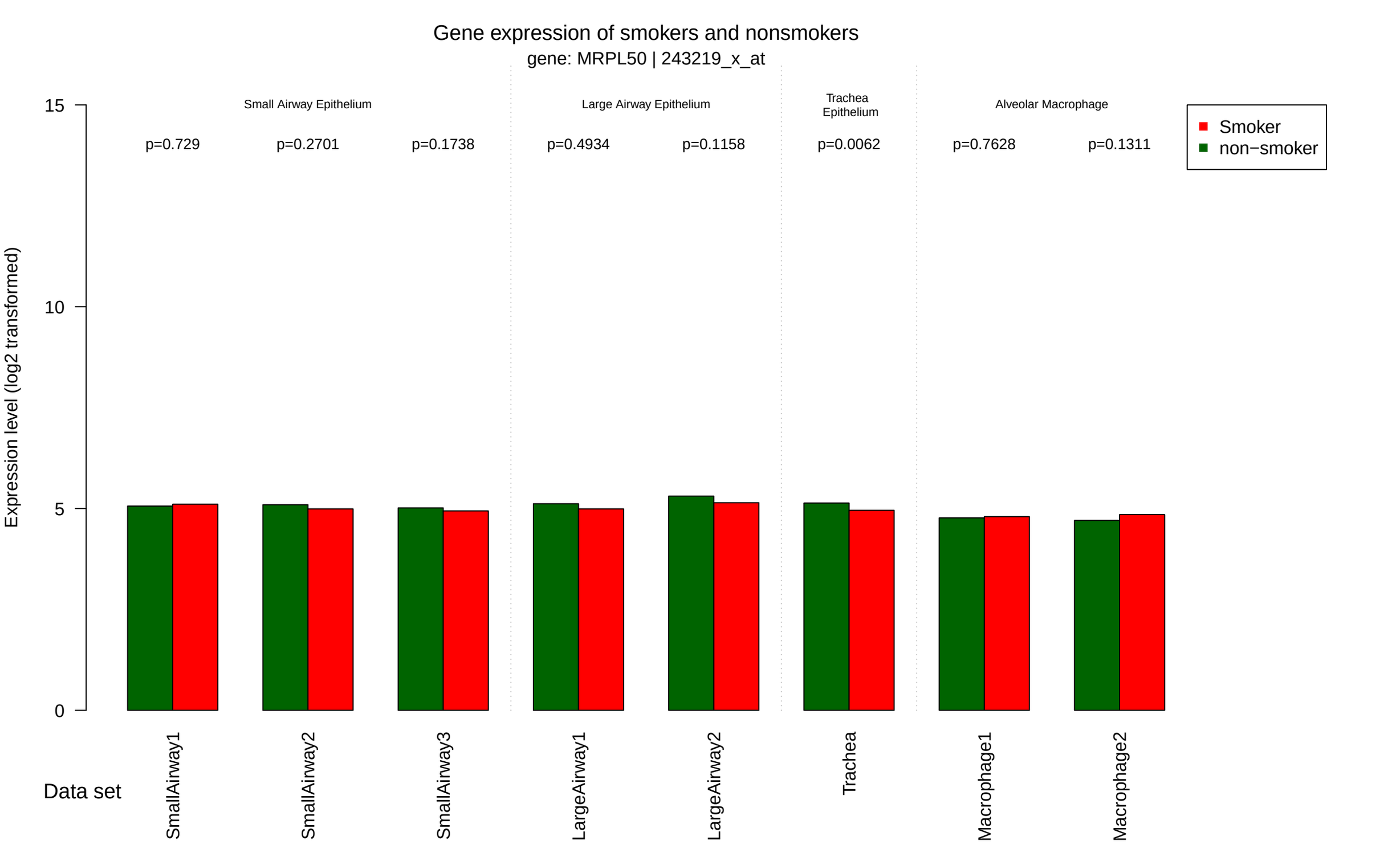
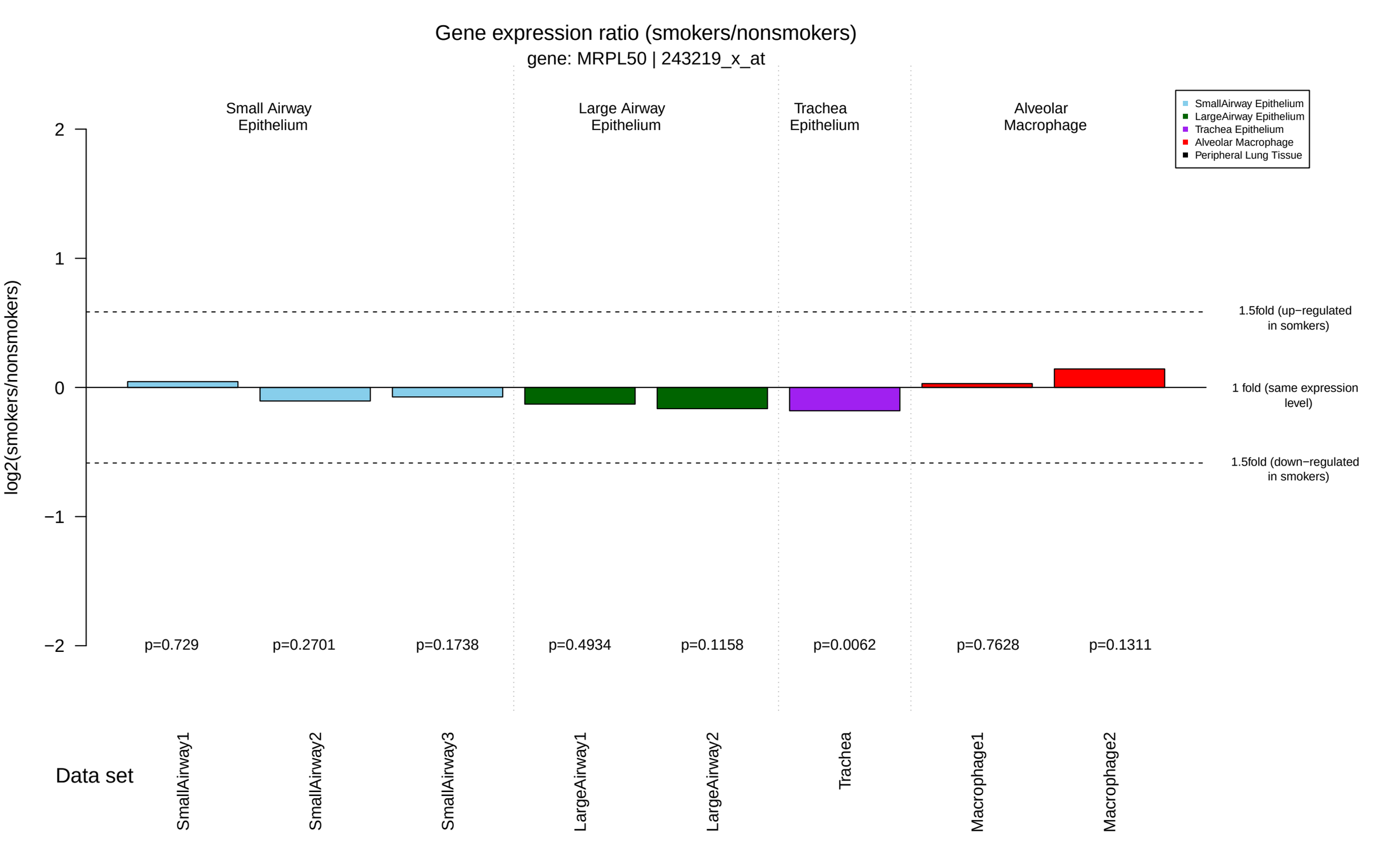
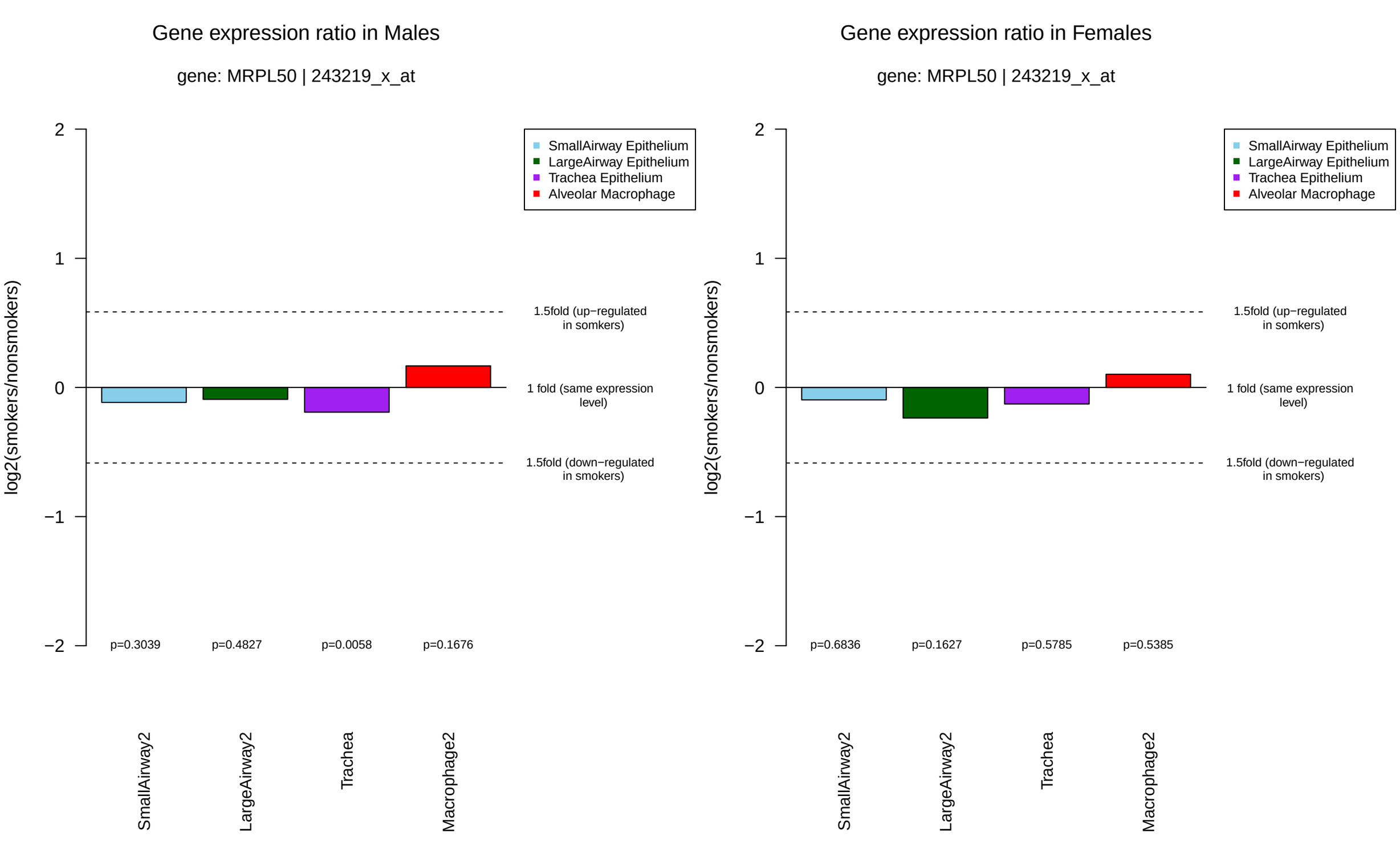
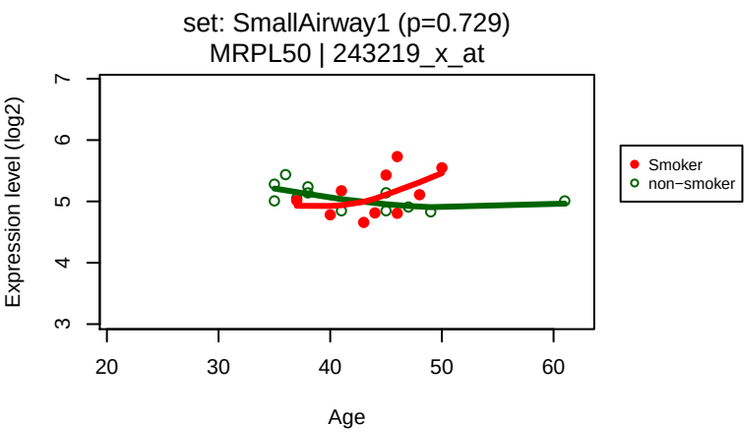
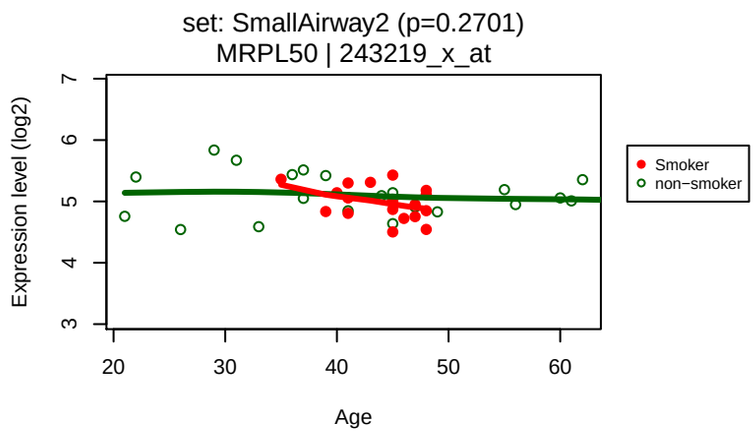
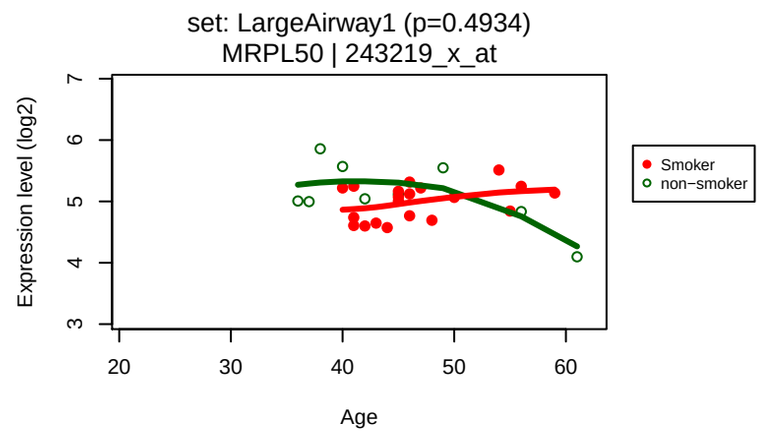

Gene expression of smokers and nonsmokers
gene: MRPL50 | 243219_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.729
p=0.2701
p=0.1738
p=0.4934
p=0.1158
p=0.0062
p=0.7628
p=0.1311
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

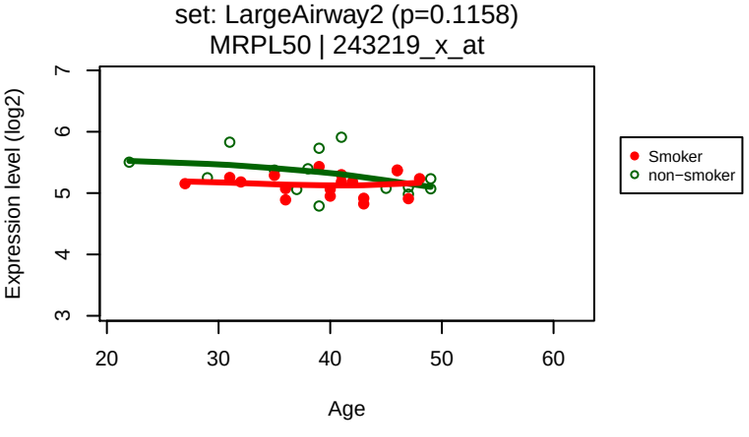

Gene expression ratio (smokers/nonsmokers)
gene: MRPL50 | 243219_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.729
p=0.2701
p=0.1738
p=0.4934
p=0.1158
p=0.0062
p=0.7628
p=0.1311
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

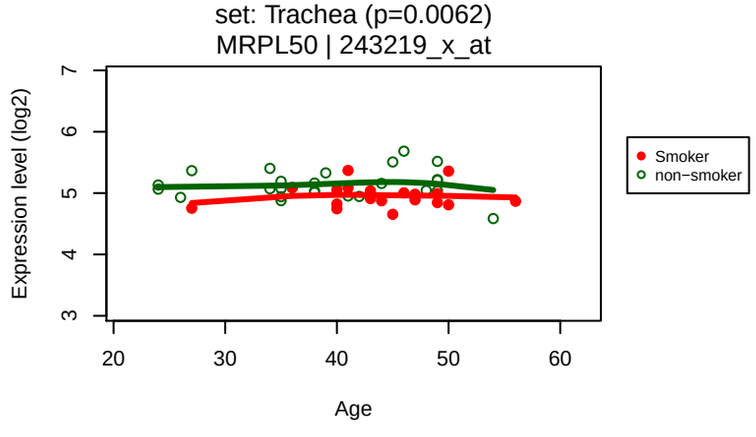

Gene expression ratio in Males
gene: MRPL50 | 243219_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3039
p=0.4827
p=0.0058
p=0.1676
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL50 | 243219_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6836
p=0.1627
p=0.5785
p=0.5385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

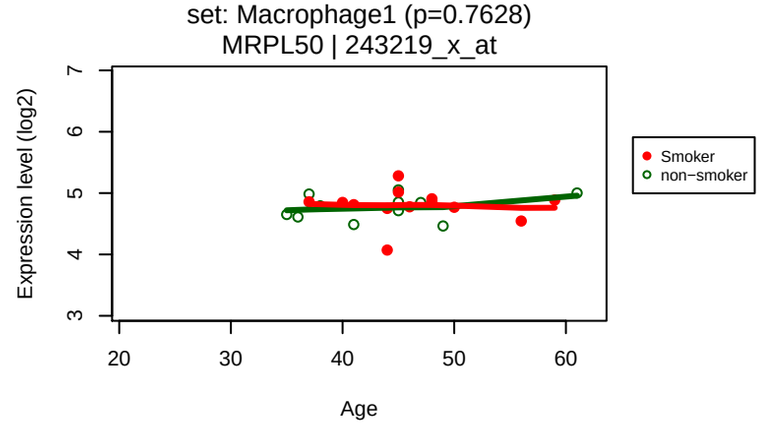

set: SmallAirway1 (p=0.729)
MRPL50 | 243219_x_at
Expression level (log2)
Age
Smoker
non−smoker

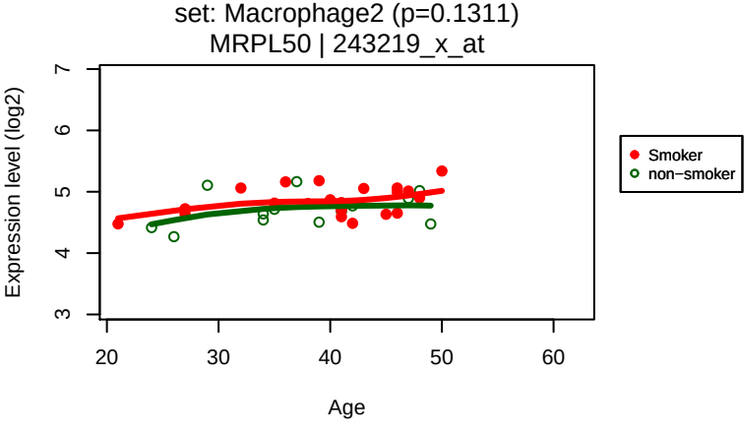

set: SmallAirway2 (p=0.2701)
MRPL50 | 243219_x_at
Expression level (log2)
Age
Smoker
non−smoker

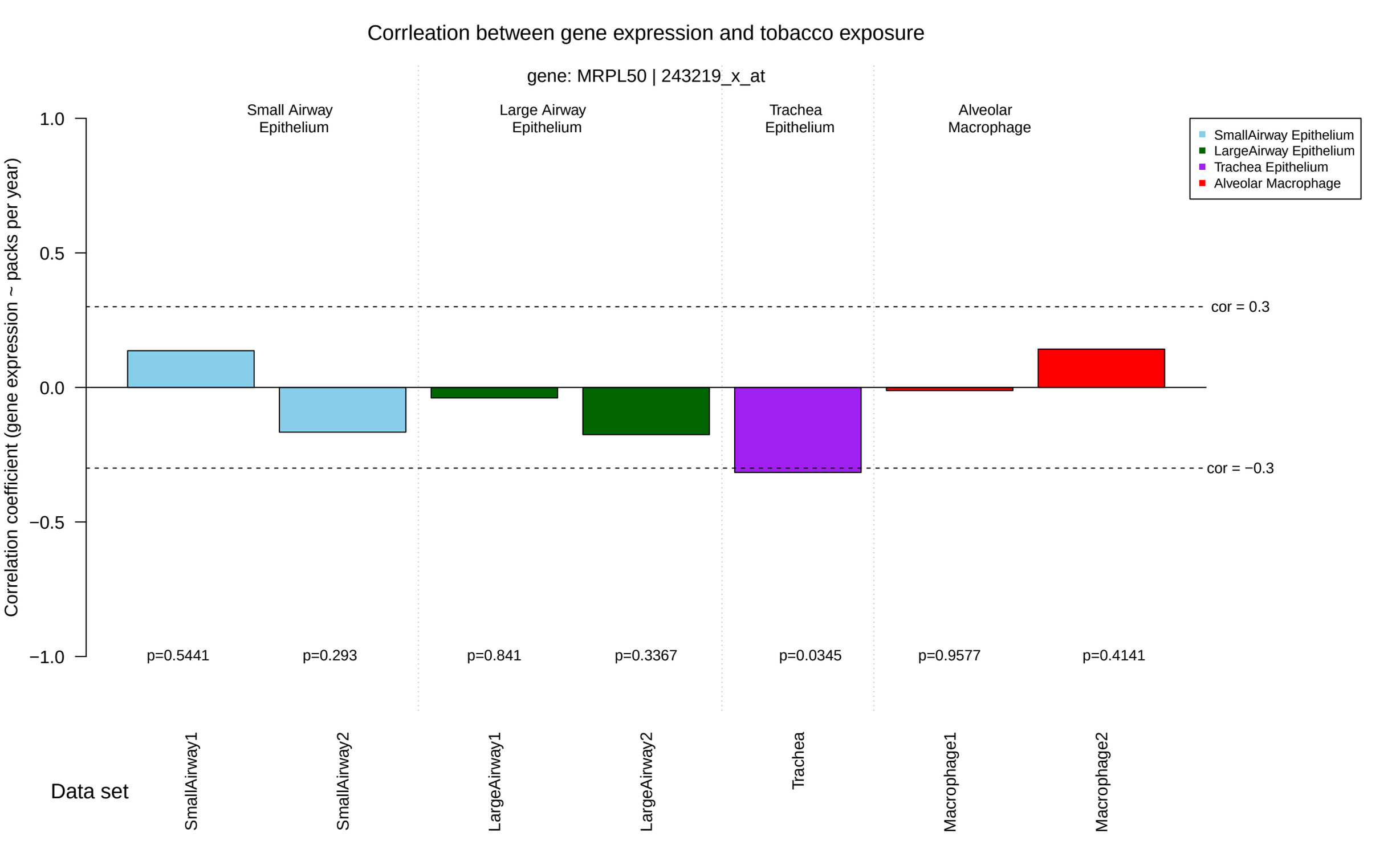

set: LargeAirway1 (p=0.4934)
MRPL50 | 243219_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1158)
MRPL50 | 243219_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0062)
MRPL50 | 243219_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7628)
MRPL50 | 243219_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1311)
MRPL50 | 243219_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL50 | 243219_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5441
p=0.293
p=0.841
p=0.3367
p=0.0345
p=0.9577
p=0.4141
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2